GRADE 5

Computational Thinking

National Level Examination

SUBJECT CODE | | | |

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **CAREFULLY CHECK YOUR NAME, CLASS, ROLL NUMBER, AND OTHER REQUIRED DETAILS ON THE OMR ANSWER SHEET.**
- **DO NOT FORGET TO SIGN THE OMR SHEET. ALSO WRITE YOUR ROLL NO. ON THE QUESTION PAPER BOOKLET.**
- **USE ONLY BLACK/BLUE BALL PEN AND HB PENCIL TO FILL THE OMR ANSWER SHEET. • ALL QUESTIONS ARE COMPULSORY AND CARRY ONE MARK EACH.**
- **SECTION A: ABSTRACT & SPATIAL THINKING (10 QUESTIONS); SECTION B : PATTERN RECOGNITION (10 QUESTIONS); SECTION C: DECOMPOSITION, NUMBERS & DATA (10QUESTIONS); SECTION D: ALGORITHMIC THINKING & LOGICAL SEQUENCING (10 QUESTIONS) SECTION E – OLYMPIAD CHALLENGE**
- **ONLY OMR ANSWER SHEETS WILL BE EVALUATED.**
- **NO NEGATIVE MARKING FOR INCORRECT .**
- **YOU MAY USE THE LAST PAGE OF THE QUESTION PAPER FOR ROUGH WORK (IF NEEDED).**
- **DIGITAL DEVICES LIKE CALCULATORS, SMART WATCHES, OR MOBILE PHONES ARE NOT ALLOWED.**
- **AFTER THE EXAM, SUBMIT BOTH THE OMR SHEET AND THE QUESTION PAPER TO THE INVIGILATOR.**

SECTION A – ABSTRACT & SPATIAL THINKING

Q. A robot starts from Point A facing North. It moves 3 steps forward, turns right, moves 4 steps, turns right, moves 3 steps, turns left and moves 2 more steps. Where is the robot now with respect to Point A?

A. 6 steps East
B. 6 steps West
C. 2 steps North
D. 4 steps East

Q2. Two identical paper sheets are joined together at one of their sides to form a square. Which of the following can definitely not be the shape of each of the paper sheets ?
A. Triangle
B. Square
C. Rectangle
D. None of these

Q3. Which figure has exactly two lines of symmetry?
A. Square
B. Rectangle
C. Circle
D. Equilateral triangle

Q4. A square sheet of paper is folded once vertically and then once horizontally. One hole is punched through the folded paper at a point that does not lie on any fold. How many holes will appear when the paper is completely opened?

A. 2
B. 3
C. 4
D. 8

5. Aarav is facing East. He makes three quarter-turns clockwise. Which direction is he facing now?
A. North
B. South
C. East
D. West

Q6. A design has rotational symmetry of order 4. What is the smallest angle through which it can be rotated to look exactly the same?

A. 45°
B. 60°
C. 90°
D. 180°

Q7. P, Q, R and S stand in a row. S is immediately to the right of P and immediately to the left of Q. R is to the right of Q. Who is second from the right?

A. P
B. Q
C. R
D. S

Q8. A child is facing South-West. After making a half-turn, which direction will the child face?

A. South-East
B. North-East
C. North-West
D. East

Q9.**Three cubes are placed one exactly above another to form a tower. How many cube faces are hidden because they touch another cube?**
A. 2
B. 3
C. 4
D. 6

Q10. A square is cut along both its diagonals. How many of the resulting triangles have the centre of the square as one of their vertices?

A. 2
B. 3
C. 4
D. 8

SECTION B – PATTERN RECOGNITION

Q11. Find the next number: 3, 7, 15, 31, ___
A. 47
B. 57
C. 63
D. 65

Q13. Which letter comes next? A, C, F, J, O, ___

A. T
B. U
C. V
D. W

Q14. What will come in place of ' ? 'in the given series ?
1@3 , #5# , 7@9 , #11# , 13@15 ,? , 19 @21

A. @17@
B. #15#
C. 16#18
D. #17#

Q15. What comes next? 5, 10, 8, 16, 14, 28, ___

A. 24
B. 26
C. 30
D. 56

Q16. The following number series is based on a pattern. One term is incorrect. Identify the incorrect term 2, 3, 6, 11, 18, 25, 38, 51, 66

A. 11
B. 51
C. 25
D. 66

Q12. Find the missing number: 2, 6, 12, 20, 30, ___
A. 36
B. 40
C. 42
D. 44

Q17. Find the next pair: AZ, BY, CX, ___

A. DV
B. DW
C. DX
D. EV

Q18. A number machine changes a two-digit number into the sum of the squares of its digits. For example: 23 → $2^2 + 3^2 = 13$. What will 52 become?

A. 25
B. 27
C. 29
D. 49

Q19. Find the next number: 81, 27, z9, 3, ___

A. 0
B. 1
C. 2
D. 6

Q20. The following pattern repeats continuously: ● ■ ▲ ◆ ● ■ ▲ ◆ ... Which symbol will be in the 27th position?

A. ●
B. ■
C. ▲
D. ◆

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SECTION C – DECOMPOSITION, NUMBERS & DATA

Q 21. Using the digits 0, 2, 5, 7 and 9 exactly once, what is the largest possible 5-digit even number?

A. 97520
B. 97502
C. 97250
D. 95720

Q 22. What is the smallest 5-digit number that can be formed using five different digits whose sum is 10?

A. 10234
B. 10324
C. 12034
D. 12340

Q 23. Three-fifths of the chocolates in a box are red. One-fourth of the red chocolates are given away. What fraction of all the chocolates are red and still left?

A. 3/20
B. 6/20
C. 9/20
D. 12/20

Q 27. A dairy farm collected 24 L of milk on Monday, 30 L on Tuesday, 18 L on Wednesday and 36 L on Thursday. The milk collected on Wednesday and Thursday is packed into 6-litre cans. How many cans are required?

A. 7
B. 8
C. 9
D. 10

Q 28. In a pictograph, ★ represents 4 books and half a ★ represents 2 books. Class A has ★★★★★. Class B has ★★★ and half a ★. How many more books does Class A have than Class B?

A. 4
B. 6
C. 8
D. 10

Q 24. Three-fourths of the students in a class is 24. How many students are there altogether?

A. 28
B. 30
C. 32
D. 36

Q 25. Three identical bottles filled with water weigh 4 kg 500 g altogether. Each empty bottle weighs 500 g. How much water is there in each bottle?

A. 500 g
B. 750 g
C. 1 kg
D. 1 kg 500 g

Q26. A tank contains 12 litres of juice. The juice is poured into bottles that each hold 750 mL. How many completely filled bottles can be made?

A. 12
B. 14
C. 16
D. 18

Q 29. A race begins at 10:47:35 a.m. and ends at 10:50:05 a.m. What is the duration of the race?

A. 2 min 20 sec
B. 2 min 30 sec
C. 2 min 40 sec
D. 3 min 30 sec

Q 30. Riya has four ₹10 coins and six ₹5 coins. She exchanges two ₹10 coins for four ₹5 coins. She then spends three ₹5 coins. Which option correctly gives the money and number of coins she has left?

A. ₹50 and 8 coins
B. ₹55 and 9 coins
C. ₹55 and 10 coins
D. ₹60 and 9 coins

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

31. A number machine follows these steps: Step 1: Multiply the number by 2; Step 2: Add 3; Step 3: Multiply the result by 2. What is the output if the input is 4?

A. 19
B. 20
C. 22
D. 24

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32. A machine multiplies an input by 3 and then adds 4. The output is 34. What was the input?

A. 8
B. 9
C. 10
D. 12

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33. Start with the number 5. Repeat the followinginstructions three times: (1) Add 2; (2) Multiply the result by 2. What is the final number?

A. 56
B. 60
C. 64
D. 68

Q 34. A robot moves 3 squares East, 2 squares North, 1 square West and 4 squares South. In which direction is its final position from its starting point?

A. North-East
B. South-East
C. South-West
D. North-West

Q 34. A robot moves 3 squares East, 2 squares North, 1 square West and 4 squares South. In which direction is its final position from its starting point?

A. North-East
B. South-East
C. South-West
D. North-West

Q 35. Alex, Jim and Sam were born on prime-numbered dates in the same month and year. Sam was born on a Friday, and the month started on a Sunday

- **The difference Between the dates of Alex and Jim's Birthday is 20**
- **Jim was born 10 days after Sam**
- **on which day of the week was Alex born ?**

A. Thursday
B. Wednesday
C. Tuesday
D. Saturdat

Q 36. Four activities B, A, D and C must be performed according to these rules: B is immediately before A; D is immediately after A; C is after D. Which activity must be performed third?

A. A
B. B
C. C
D. D

SAMPLE PAPER

Q 37. A number follows this rule: If it is even, divide it by 2. If it is odd, add 5. Starting with 18, apply the rule four times. What is the final number?
A. 7
B. 9
C. 12
D. 14

Q 38. In the following series , which of these English alphabets would appear first alongside a negative number?A165 , B153 , C141, D129, ...
A.M
B.N
C. O
D. P

Q 39. A student wants a robot to: (1) Move 4 steps forward; (2) Turn right; (3) Move 2 steps forward; (4) Turn left; (5) Move 1 step forward. The student accidentally writes Turn Left in Step 2. Which is the best way to debug the algorithm?
A. Remove Step 1
B. Change Step 2 from Left to Right
C. Change Step 3 to 4 steps
D. Repeat the whole program twice

Q 40. A robot can jump either 2 squares or 3 squares in one move. What is the minimum number of jumps needed to move exactly 12 squares?
A. 3
B. 4
C. 5
D. 6

SECTION E – OLYMPIAD CHALLENGE

Q 41. A 5-digit secret code uses the digits 1, 2, 3, 4 and 5 exactly once. The third digit is 5. The last digit is even. The sum of the first two digits is 5. The first digit is greater than the second digit. The fourth digit is greater than the last digit. What is the code?

A. 41532
B. 32514
C. 14532
D. 23514

Q 42. A tank is 3/8 full and contains 9 litres of water. Seven more litres are added. What fraction of the tank is now filled?

A. 1/2
B. 5/8
C. 2/3
D. 3/4

Q. 43. A robot starts facing North. It turns according to this repeating pattern: 90° clockwise, 180° clockwise, 90° clockwise, 180° clockwise... Which direction will it face after the fifth turn?

A. North
B. South
C. East
D. West

Q44. Five children — A, B, C, D and E — are of different ages. A is older than B. C is older than A. D is older than C. E is younger than B. Who is the third oldest?

A. A
B. B
C. C
D. D

Q 45. A secret 3-digit number has different digits. Its hundreds digit is twice its units digit. Its tens digit is one less than its hundreds digit. The sum of its digits is 14. What is the number?

A. 432
B. 653
C. 764
D. 874

Q46. A robot has to travel from one corner of a grid to another using exactly 2 moves Right and 2 moves Up. It may perform these four moves in any order. How many different shortest routes are possible?

A. 4
B. 5
C. 6
D. 8

Q 47. Two apples and one orange together weigh 500 g. One apple and one orange together weigh 300 g. What will 3 apples and 2 oranges weigh?

A. 600 g
B. 700 g
C. 800 g
D. 900 g

Q 48. Study the number machine: 1 → 4, 2 → 7, 3 → 10, 4 → 13. If the same rule is followed, what will be the output for 5?

A. 14
B. 15
C. 16
D. 17

Q 49. On a map, the library is 3 blocks East and 2 blocks North of the school. The park is 1 block West and 4 blocks North of the library. Where is the park relative to the school?

A. 2 blocks East and 6 blocks North
B. 4 blocks East and 6 blocks North
C. 2 blocks West and 2 blocks North
D. 4 blocks East and 2 blocks South

Q 50. A number machine follows this rule: If the number is even, divide it by 2. If the number is odd, multiply it by 3 and add 1. Starting with 10, apply the rule five times. What is the final number?

A. 1
B. 2
C. 4
D. 8

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